

Index Option Return Forecasting using Machine Learning with Equivalent Expectation Measure-Based Predictors

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Abstract

This paper uses 10 types of machine learning approaches to predict future holding-to-maturity S&P 500 index option returns from 1998 to 2021 for various option moneynesses and maturities. To assess the importance of economic theory in machine learning, we incorporate option return predictors based on the Equivalent Expectation Measure (EEM) Theory (see [Nawalkha and Zhuo \(2022\)](#)), which can give analytical solutions of expected future option prices and expected option returns over any finite holding period. We find that including the EEM-based predictors significantly enhances the out-of-sample performance of most machine learning models. In addition, compared with traditional stock characteristics and macroeconomic factors, the EEM-based predictors are superior on forecast accuracy. Finally, we also confirm that nonlinear machine learning methods outperform linear ones in predicting index option returns.

Keywords: Option return, machine learning, equivalent expectation measure

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1. Introduction

Financial markets are characterized by a low signal-to-noise ratio, presenting significant challenges in asset pricing. Nagel (2021) specifically points out the limitations of relying solely on data-driven machine learning frameworks in this context. He argues for a more structured approach that integrates economic reasoning and prior knowledge into machine learning methodologies. In response to this call, our research advances the integration of economic structural models with machine learning techniques, focusing particularly on option index returns—a sector where strong structural modeling is already established, yet machine learning applications remain relatively unexplored. By blending rigorous economic theories with sophisticated machine learning algorithms, our approach not only adheres to the theoretical underpinnings of economic frameworks but also enhances the predictive accuracy of asset pricing models. This novel methodology offers a robust and theoretically informed solution, tackling the complexities of financial markets with enhanced precision, especially in the domain of option index returns.

Over the past decade, option markets have significantly increased in importance, as evidenced by the more than tenfold rise in options trading volume on global exchanges—from \$9.43 billion contracts in 2013 to \$108.19 billion contracts in 2023.¹ Given its significance, understanding what drives and predicts option returns is a crucial research question. There is fast-growing literature on the determinants and prediction of option returns, see, e.g., Christoffersen et al. (2018), Hu and Jacobs (2020), Hu and Liu (2022), Zhan et al. (2022), Heston et al. (2022), Goyal and Saretto (2022), Büchner and Kelly (2022), Heston et al. (2023). Among them, Bali et al. (2023) provide a comparative analysis of machine learning methods for predicting individual option returns. Despite these advancements, the potential of machine learning to improve the predictability of index option returns remains an open question.

This paper explores forecasting S&P 500 index option returns using various machine learning methods and diverse predictors. This research is critically important for multiple

¹The data is from the Futures Industry Association (FIA)’s annual statistical review, see from <https://www.fia.org/fia/etd-tracker>.

reasons. Firstly, in 2023, the trading volume of equity index options on global exchanges surged to \$87.04 billion contracts. This volume not only represents 80% of all options trading worldwide but is also nearly ten times the volume of individual options trading, which stood at \$8.92 billion. Secondly, unlike individual options—which are influenced primarily by specific stock movements, company-specific events, and idiosyncratic risks—index options capture broader market trends and sentiments, benefiting from significantly higher liquidity and trading volume. Therefore, index option returns might demonstrate distinct patterns that are different from those of individual options, offering deeper insights into systemic risks, market sentiment, and broader economic indicators. Moreover, machine learning methods could reveal unique capabilities in forecasting returns of index options compared to individual options. This paper will elucidate the patterns in S&P 500 index option returns and evaluate the effectiveness of different machine learning approaches in predicting these returns.

There is fast-growing literature that introduces machine learning methods to asset pricing applications, in which a variety of linear and nonlinear methods are used in *reduced-form* cross-sectional and time-series forecasting of different asset returns, such as stock returns (Gu et al. (2020), Leippold et al. (2022)), treasury securities (Bianchi et al. (2021)), exchange rates (Filippou et al. (2020)), mutual funds (DeMiguel et al. (2021)), and individual options (Bali et al. (2023)). In this paper, we are in favor of Nagel (2021)’s insights and leverage the prior knowledge embedded in *structural-form* option models. This foundational understanding guides our identification of valuable predictors and informs the structure of our learning algorithm, enhancing both its relevance and effectiveness.

Nawalkha and Zhuo (2022) introduced the Equivalent Expectation Measure (EEM) theory, which provides analytical solutions for the expected future prices and returns of options over any finite period. The EEM framework systematically integrates structural economic theories with empirical data, offering a unique analytical advantage. This theory offers a comprehensive economic framework that can predict finite-horizon returns under various structural option models, including the classical Black-Scholes model

Black and Scholes (1973), the stochastic volatility model Heston (1993), the affine jump-diffusion model Bakshi and Madan (2000), Duffie et al. (2000), and the infinite-activity Lévy model Carr et al. (2002). Utilizing the prior knowledge from the EEM theory, this paper categorizes option return predictors into two types:

- *EEM-based determining predictors*: These predictors derive from the input variables of structural option models. They are theoretical determinants of expected option returns under EEM theory and serve as potential empirical predictors of realized option returns.
- *EEM-based resulting predictors*: These predictors are based on the output variables of structural option models, which predict expected future option prices and returns. These are the theoretical expectations utilized when applying machine learning techniques.

Incorporating these *EEM-based predictors* to forecast index option returns using machine learning methods prompts two critical inquiries that this paper will address: Do these EEM-based predictors offer incremental information for predicting index option returns within machine learning frameworks? Furthermore, how do these EEM-based predictors perform relative to traditional predictors like option and stock characteristics?

To answer these questions, this paper uses S&P 500 index options data from OptionMetrics IvyDB over the period from 1998 to 2021, and employs 10 types of machine learning algorithms to predict future holding-to-maturity excess returns of S&P 500 index options for various option moneynesses and maturities. Following Gu et al. (2020) and Bali et al. (2023), the machine learning models encompass both the dimensionality reduction type (e.g., *Principal Component Regression*, *Partial Least Squares*) and the penalized regression type (e.g., *Lasso*, *Ridge*, *Elastic-Net*) in linear models, as well as the tree-based type (e.g., *GBDT*, *Random Forest*) and Neural Network type in nonlinear models. The key predictors/inputs in machine learning methods are 8 EEM-based determining predictors and 2 EEM-based resulting predictors produced by the EEM theory of Nawalkha and Zhuo (2022) under the Black and Scholes (1973) structural option model; the benchmark predictors/inputs in machine learning methods include 5 standard option

characteristics (e.g., implied volatility, implied premium) and 7 aggregated stock market characteristics and macroeconomic predictors (e.g., dividend price ratio, stock variance, term spread) from [Welch and Goyal \(2008\)](#). To evaluate the impact of different predictors on the prediction of index option returns, we implement each machine learning method under three distinct scenarios. First, we include only benchmark predictors. Then, we introduce EEM-based determining predictors alongside the benchmark set. Finally, we incorporate all benchmark predictors, EEM-based determining predictors, and EEM-based resulting predictors. This gradual approach allows us to assess the incremental information contributed by EEM-based predictors. Following the methodology outlined by [Gu et al. \(2020\)](#), we employ a recursive validation strategy for each model and utilize out-of-sample R^2 -statistics (against a baseline of zero excess return) and mean squared errors (MSE) as performance metrics. Further, we quantify the importance of each predictor by calculating the decrease in R^2 when the predictor’s values are set to zero within each training sample, averaging these reductions to derive a comprehensive importance measure for each predictor.

Our investigation yields three primary findings that significantly advance the field of financial predictive modeling. Firstly, the integration of EEM-based predictors into a diverse array of machine learning models, with the exceptions of Partial Least Squares (PLS) and General Linear Models (GLM), markedly enhances their predictive accuracy. This improvement is particularly notable during economic crisis periods. The inclusion of EEM-based resulting predictors increases the average out-of-sample R^2 by as much as 4.39% across various machine learning models, with the greatest enhancement observed in the Gradient Boosting Decision Tree (GBDT) models. In contrast, Random Forest models exhibit a slight decline in performance. We attribute this to the complex nature of EEM-based predictors, which may not be adequately captured by models functioning as “weak learners”. Secondly, our analysis of predictor importance reveals that EEM-based predictors significantly contribute to model effectiveness. The exclusion of these predictors consistently results in a substantial reduction in R^2 across all tested machine learning models. Remarkably, in the application of the Lasso model, only five

variables are selected, all of which are EEM-based predictors, thereby underscoring their pivotal role. Among traditional stock characteristics, net equity expansion emerges as the most influential predictor, likely due to its strong correlation with market expectations. Thirdly, consistent with the findings of Gu et al. (2020) and Bali (2023), our study confirms that machine learning methods generally surpass the traditional Ordinary Least Squares (OLS) approach in predictive accuracy. This superiority extends to comparisons between nonlinear and linear algorithms, with nonlinear methods demonstrating a significant edge, a trend that persists even during periods of economic crisis. These findings underscore the value of integrating advanced economic theories via EEM-based predictors into contemporary machine learning frameworks, highlighting a significant progression in the capabilities of financial predictive models.

Our research makes several key contributions to the literature on financial predictive modeling, particularly in the integration of economic theories with machine learning algorithms for the forecasting of index option returns.

First Contribution: Our study is notable for its application of the EEM theory, as proposed by [Nawalkha and Zhuo \(2022\)](#), to the empirical prediction of future index option returns. This theory, which establishes hybrid probability measures, provides a theoretical framework for anticipating the physical expectation of future prices and serves as a pricing measure beyond the horizon. Prior applications, such as those by [Huang and Shi \(2021\)](#), have effectively used EEM-derived indicators to predict corporate bond returns. Extending this to index options, our research integrates EEM-based structural models with machine learning techniques to enhance the precision of financial models. This approach, which merges theoretical economic constructs with empirical methodologies, represents a methodology not yet extensively explored in existing literature.

Second Contribution: We introduce and validate new EEM-based predictors derived from structural-form option models. While extensive research, such as that by [Goodman et al. \(2018\)](#) and [Christoffersen et al. \(2018\)](#), has focused on the empirical, reduced-form determinants of option returns, our approach utilizes a structural-form basis. We demonstrate that these EEM-based predictors offer additional predictive value

over traditional methods, thus providing a viable alternative for accurately forecasting future index option returns.

Third Contribution: Our work includes a comprehensive examination of various machine learning methodologies for forecasting S&P 500 index option returns. Previous studies, such as those by [Bali et al. \(2023\)](#) and [Büchner and Kelly \(2022\)](#), have applied machine learning to individual stock options or developed specific models like the three-factor IPCA for index options. Our research extends these analyses by methodically assessing the efficacy of linear versus nonlinear algorithms. Our findings confirm the enhanced performance of nonlinear algorithms in detecting complex patterns in option returns, thereby supporting and expanding the conclusions of prior research, e.g., [Gu et al. \(2020\)](#), [Leippold et al. \(2022\)](#), [Bianchi et al. \(2021\)](#), and [Bali et al. \(2023\)](#).

The remainder of the paper is organized as follows. Section 2 briefly introduces the equivalent expectation measure theory and constructs the EEM-based predictors. Section 3 presents our methodology for index option return prediction using machine learning methods and the corresponding performance valuation metrics. Section 4 provides the prediction comparison results of various machine learning methods and analyzes the importance of predictors. Section 5 concludes the paper.

2. The EEM Theory and the EEM-Based Predictors

The equivalent expectation measure (EEM) theory proposed by [Nawalkha and Zhuo \(2022\)](#) can be used to obtain the analytical solutions of expected future prices and expected returns of a variety of contingent claims over any holding horizon. Such contingent claims include not only financial derivatives (e.g., options and interest rate derivatives), but also fixed-income securities (e.g., treasury bonds and corporate bonds). In this section, we first briefly introduce its application under a classical structural option model of [Black and Scholes \(1973\)](#), then derive the corresponding EEM-based predictors, which will be used as the key predictors in our later machine learning implementations.

2.1. The EEM theory under the Black-Scholes model

Under the [Black and Scholes \(1973\)](#) model, the geometric Brownian motion for the stock price process is given as follows:

$$\frac{dS_s}{S_s} = \mu ds + \sigma dW_s^{\mathbb{P}},$$

where μ is the drift, σ is the volatility, and $W^{\mathbb{P}}$ is the Brownian motion under the physical measure $\mathbb{P}$.

Given an investment horizon $H(\leq T)$ for option return calculation and the above standard Brownian motion $W^{\mathbb{P}}$ under $\mathbb{P}$, as in [Nawalkha and Zhuo \(2022\)](#), the standard Brownian motion under the equivalent expectation measure $\mathbb{R}$ is derived as follows:

$$W_s^{\mathbb{R}} = W_s^{\mathbb{P}} + \int_0^s \mathbb{1}_{\{u \geq H\}} \gamma du,$$

where γ is the market price of risk and $\mathbb{1}_{\{\cdot\}}$ is an indicator function that equals one if the condition is satisfied (and zero otherwise). It follows that the stock price process under the $\mathbb{R}$ measure is given by

$$\frac{dS_s}{S_s} = (r + \sigma\gamma\mathbb{1}_{\{s < H\}}) ds + \sigma dW_s^{\mathbb{R}} \tag{1}$$

$$= (\mu\mathbb{1}_{\{s < H\}} + r\mathbb{1}_{\{s \geq H\}}) ds + \sigma dW_s^{\mathbb{R}}, \tag{1a}$$

where r is the risk-free rate, and $\gamma = (\mu - r)/\sigma$.

A highly intuitive property of the equivalent expectation measure $\mathbb{R}$ is that any stochastic process under this measure, such as the stock (index) price process, is the physical stochastic process until before the horizon H , but it becomes the risk-neutral process on or after the horizon H .

Consider a European call option C on the above stock S , maturing at time T with a strike price K . If one investor buys this call option at current time t and plans to hold it until a future time H with $t \leq H \leq T$, as in [Nawalkha and Zhuo \(2022\)](#), we can obtain

the time- t expectation of the call option at a future time H as follows:

$$\mathbb{E}_t[C_H] = \mathbb{E}_t^{\mathbb{R}} \left[e^{-r(T-H)} (S_T - K)^+ \right], \quad (2)$$

where C_H is the price of the call option at time H . Since equation 1 implies that S_T is lognormally distributed under the $\mathbb{R}$ measure, the expected call price can be solved easily and is given as

$$\mathbb{E}_t[C_H] = S_t e^{\mu(H-t)} \mathcal{N}(\hat{d}_1) - K e^{-r(T-H)} \mathcal{N}(\hat{d}_2), \quad (3)$$

where $\mathcal{N}(\cdot)$ is the standard normal cumulative distribution function, and

$$\begin{aligned} \hat{d}_1 &= \frac{\ln(S_t/K) + \mu(H-t) + r(T-H) + \frac{1}{2}\sigma^2(T-t)}{\sigma\sqrt{T-t}}, \\ \hat{d}_2 &= \frac{\ln(S_t/K) + \mu(H-t) + r(T-H) - \frac{1}{2}\sigma^2(T-t)}{\sigma\sqrt{T-t}}. \end{aligned}$$

When $H = t$, the above formula reduces to that classical Black-Scholes formula in Black and Scholes (1973). With the above analytical solution of expected future option price in (3), we can then obtain the analytical solution of expected option excess return over a holding period from time t to H as follows:

$$\mathbb{E}_t[r_H^C] = \frac{\mathbb{E}_t[C_H]}{C_t} - 1 - r, \quad (4)$$

where C_t is the current option price derived by the classical Black-Scholes formula, or by equation (3) with $H = t$.

Similarly, for a put option P , we can also derive the corresponding analytical solutions of expected future price $\mathbb{E}_t[P_H]$ and expected excess return $\mathbb{E}_t[r_H^P]$.

2.2. The EEM-based predictors

Note that six variables determine the model-implied expected option price/return in equations (3) and (4) in a structural-form way, including the underlying stock (index)'s current price S_t , the option strike price K , the expected return of stock (in-

dex) μ , the risk-free rate r , the stock (index) volatility σ , the option maturity $T - t$, and the option holding period $H - t$. Moreover, they predict expected call and put returns in different fashions, hence a dummy variable that decides call or put will be needed. We then refer these 8 variables as the *EEM-based determining predictors*, and label them as “underlying_price”, “strike_price”, “underlying_ret”, “risk_free”, “underlying_vol”, “holding_period”, “option_maturity”, and “if_call”, respectively. Notably, these EEM-based determining predictors can generally count as stock and option characteristics, but under the EEM theory, they are the input variables that decide the structural model-implied expected option price/return with a quite complex nonlinear relation among them. Hence they are more valuable than other standard stock and option characteristics and are taken out as one type of EEM-based predictors.

On the other hand, the expected future option price in equation (3) and the expected option return in equation (4) are the derivations of the EEM theory, we hence refer to these two variables as the *EEM-based resulting predictors*, and label them as “EEM_price” and “EEM_ret”, respectively. They are the agents of sophisticated structural-form information hidden in the above EEM-based determining predictors, we think they provide the most valuable information in predicting future option returns.

We collectively refer to the above two types of predictors as *EEM-based predictors*, which will be used as the key predictors in our later machine learning implementations. All of the above-mentioned EEM-based predictors can be directly measured or estimated by using historical stock (index) data.

3. Methodology and Performance Evaluation

This section first presents our basic framework for index option return prediction using machine learning methods, then describes the corresponding performance valuation standards and recursive method of machine learning.

For an option indexed by i ($i = 1, \dots, N_t$) and date by t ($t = 1, \dots, T_i$), T_i is the option

i 's maturing date, we define its realized holding-to-maturity excess return as

$$r_{i,T_i} = \frac{O_{i,T_i}}{O_{i,t}} - 1 - r_{f,t}, \quad (5)$$

where $O_{i,t}$ is the option i 's price at date t , O_{i,T_i} is the option i 's price at its maturing date T_i , $r_{f,t}$ is the yield to maturity of U.S. treasury bonds corresponding to the option i 's maturity at date t . We then express the option's realized excess return as an additive prediction error model as follows:

$$r_{i,T_i} = \mathbb{E}_t[r_{i,T_i}] + \epsilon_{i,T_i}, \quad (6)$$

where

$$\mathbb{E}_t[r_{i,T_i}] = f^*(X_{i,t}), \quad (7)$$

where $f^*(\cdot)$ is a flexible function of predictors $X_{i,t}$, i.e., a mapping set by various machine learning methods. The predictors $X_{i,t}$ include the EEM-based predictors defined in section 2.2, and/or the benchmark option and stock characteristics. For the benchmark option characteristics, we use 5 predictors consisting of the implied volatility from option price (“`impl_vol`”), the option's implied premium (“`impl_premium`”), the option's theoretical Black-Sholes price (“`BS_price`”), the option's Delta value (“`option_delta`”), and the option's moneyness (“`option_moneyness`”). For the benchmark stock characteristics, we borrow 7 aggregated stock market characteristics and macroeconomic predictors from Welch and Goyal (2008), which include dividend price ratio (“`dp`”), earnings price ratio (“`ep`”), book-to-market ratio (“`bm`”), net equity expansion (“`ntis`”), term spread (“`tms`”), default yield spread (“`dfy`”), and stock variance (“`svar`”).

3.1. Machine learning methods

Following Gu et al. (2020) and Bali et al. (2023), we use a variety of linear and nonlinear machine learning methods to train the mapping function $f^*(\cdot)$ in our empirical analysis.

All machine learning methods and their respective main features are shown in Table 1. For linear models, we consider OLS, Principal Component Regression (PCR), Partial Least Squares (PLS), Generalized Linear Model (GLM), Lasso, Ridge, and Elastic-Net. Linear models are widely used in traditional econometrics but are unable to discover nonlinear relationships between variables. For nonlinear models, we consider GBDT, Random Forest, and Neural Networks with 1-5 hidden layers (NN1-NN5). Nonlinear models have a stronger ability to fit complex data, but they are also prone to overfitting. Therefore, it requires the adjustment of complex hyperparameters to achieve optimal performance. For technical details of different machine learning methods, we refer to [Gu et al. \(2020\)](#), [Hastie et al. \(2009\)](#), [Parikh and Boyd \(2014\)](#), [Polson et al. \(2015\)](#), and [Goodfellow et al. \(2016\)](#).

Table 1: Summary of machine learning methods

Type	Machine Learning Area	Model	Features
Linear	-	OLS	Using least squares method to fit a linear model, suitable for predicting continuous target variables.
	Dimension Reduction	PCR	Combining principal component analysis and linear regression to reduce dimensionality and address multicollinearity.
		PLS	Similar to PCR, but considering the target variable during modeling, suitable for high-dimensional datasets.
	Regularization	Lasso	Utilizing L1 regularization to sparsify the model, aiding in feature selection and model interpretability.
		Ridge	Employing L2 regularization to prevent overfitting, suitable for addressing multicollinearity issues.
		Elastic Net	Balancing L1 and L2 regularization, combining characteristics of Lasso and Ridge, suitable for complex datasets.
		GLM	Using a link function to connect predictors to the response variable, and allows the variance of measurements to depend on their predicted value.
Nonlinear	Boosting Tree	GBDT	Using gradient boosting decision trees, known for high efficiency and performance, suitable for large-scale datasets.
	Bagging Tree	Random Forest	An ensemble method based on multiple decision trees, characterized by robustness and good generalization.
	Neural Networks	NN1-NN5	Utilizing neurons and multi-layer neural networks for model training, suitable for complex nonlinear relationship modeling.

Notes: This table categorizes various machine learning models into linear and nonlinear types and outlines their main features, and suitability for particular tasks.

3.2. The standard of evaluating models

As in [Gu et al. \(2020\)](#) and [Bali et al. \(2023\)](#), we use the out-of-sample R^2 -statistic to measure the predictive ability of machine learning models on option returns, defined as follows:

$$R_{OS}^2 = 1 - \frac{\sum_{(i,t) \in \mathcal{T}_3} (r_{i,T_i} - \hat{r}_{i,T_i})^2}{\sum_{(i,t) \in \mathcal{T}_3} r_{i,T_i}^2}, \quad (8)$$

where $\mathcal{T}_3$ is the testing sample,² $\hat{r}_{i,T_i}$ is the index option excess return forecast of a given machine learning model. R_{OS}^2 measures the reduction in the mean squared forecast error (MSFE) compared to a naive benchmark of zero excess returns for all index options. Furthermore, our index option dataset is in daily frequency, hence our R_{OS}^2 is daily-based.

At the same time, we use mean squared error (MSE) as another evaluation measure, which is the average squared prediction error of a forecasting model against actual observations, defined by

$$\text{MSE} = \frac{1}{N} \sum_{(i,t) \in \mathcal{T}_3} (r_{i,T_i} - \hat{r}_{i,T_i})^2. \quad (9)$$

3.3. Recursive method of machine learning

The recursive method is widely used in the machine learning return prediction of stocks, bonds, and financial derivatives. Figure 1 illustrates the corresponding recursive cross-validation scheme in this paper. According to [Gu et al. \(2020\)](#) and [Bali et al. \(2023\)](#), given a sample period year t ($2003 \leq t \leq 2021$ in our sample), the training sample is based on observations from a fixed starting year 1998 to the year $t-3$, and the validation sample would be based on observations from year $t-2$ to year $t-1$. Then the selected model is re-estimated using all data through the validation sample, and out-of-sample forecasts for year t are generated. One iteration is finished.

Then we move to the next iteration at year $t+1$, in which the entire training, validation, and testing processes are repeated again. The training sample uses observations from year

² $\mathcal{T}_1$ is the training sample, $\mathcal{T}_2$ is the validating sample. In-sample R^2 -statistic, denoted as R_{IS}^2 , uses the same formula in equation (8) but under the training sample $\mathcal{T}_1$.

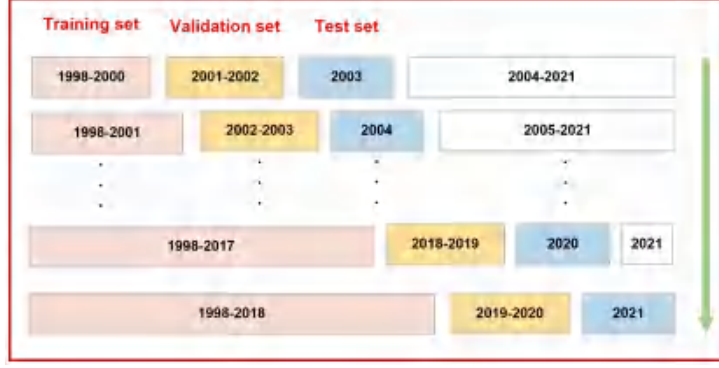


Figure 1: Recursive method of machine learning

Notes: This figure shows the recursive method we use in machine learning methods similar to [Gu et al. \(2020\)](#), which corresponds to the case of an expanding training window. The training sample is from 1998-2000 in the first year and expands for one year in every iteration. The validation sample is fixed for the next three years of the training sample and moves with iterations. The testing sample is the year after the validation sample, moving from 2003 to 2021, which takes 19 times recursion.

1998 to $t - 2$, validation sample uses observations from year $t - 1$ to year t . The selected model is estimated through the training sample and re-estimated through the validation sample, then is used to produce out-of-sample forecasts for the year $t + 1$. This recursion iterates until the last out-of-sample forecast generated for the year 2021.

All machine learning model hyperparameters are selected using the randomized search method. Because validation is re-conducted in each iteration, in every round, the model needs to be retrained. The optimal hyperparameters are ultimately determined based on the optimal R^2 in the validation sample. It is important to note that the selected model can change throughout the iteration. This is beneficial if there is suspicion of structural instability in the data.

In each iteration, we obtain the current model's R^2 and MSE for the part of the dataset used so far. After 19 times of iterations, we calculate the average R^2 and average MSE from these 19 results as evaluation performance for the current model. Finally, we summarize the results of all models and analyze them according to the model categories.

4. Predicting Index Option Returns

4.1. Data and summary statistics

We use S&P 500 index options daily data from OptionMetrics IvyDB over the period from 1998 to 2021. The descriptive statistics of our main variables are shown in Table 2. To deal with the missing values, we use the closing price of the previous trading day. For missing historical volatility and risk-free rates, we fill them using the interpolation method. In terms of outlier processing, we remove data with a quantile greater than 99% in the realized excess return and EEM-based expected return from the entire sample to deal with the impact of extreme events such as the 2008 economic crisis on model predictions.

In Figure 2, we show a graph of realized excess return and EEM-based expected return. realized excess return is our predicted variable, along with a variation in standard deviation. And EEM-based expected return is constructed by a structured model. Both of them in the graph are changing with delta. We can see that the realized excess return of puts increases as the absolute value of the delta increases. For calls, when the maturities are within 60 days, the realized excess return increases as the absolute value of the delta increases. Beyond 60 days, it first increases and then decreases. The volatility of realized excess return decreases as the absolute value of delta increases. Furthermore, the longer the maturities, the greater the degree of increase. For EEM-based expected return, the difference between realized excess return and it continues to increase as the absolute value of delta decreases seen from the full sample.

We find that our variable can achieve a good approximation of put options with maturity within 60 days. However there is a big gap between the expected return and the realized return of the call option due to the limitation of the Black-Scholes model, which only allows constant volatility and always yields positive expected returns for call options. Considering the existence of volatility risk and jump risk, we will incorporate the stochastic volatility (SV) model (see [Heston \(1993\)](#)), the stochastic volatility with jumps in price (SVJ) model (see [Pan \(2002\)](#), [Bates \(2000\)](#)) and the stochastic volatility with jumps in price and variance (SVJJ) model (see [Duffie et al. \(2000\)](#), [Eraker et al. \(2003\)](#))

Table 2: Summary statistics

	Mean	Sd	Min	Q10	Q25	Median	Q75	Q90	Max	Skew	Kurt
A: All Options:($n = 2, 108, 371$)											
Realized excess return	-0.23	1.15	-1.14	-1.02	-1.00	-0.99	0.35	1.30	5.85	1.86	3.94
EEM-based expected return	0.86	3.14	-1.13	-1.00	-0.83	-0.13	1.10	3.49	26.95	3.86	19.16
Option maturity	238.12	214.89	10.00	30.00	91.00	152.00	365.00	547.00	730.00	1.10	0.07
Option moneyness	1.01	0.13	0.41	0.86	0.94	1.01	1.06	1.14	3.05	0.49	3.95
Implied volatility	0.20	0.07	0.03	0.12	0.14	0.19	0.23	0.29	1.28	1.43	4.59
Underlying volatility	0.17	0.09	0.02	0.10	0.11	0.15	0.21	0.27	1.37	2.45	10.76
Option delta	0.00	0.56	-0.90	-0.75	-0.50	-0.10	0.50	0.75	0.90	0.01	-1.37
B: Call Options:($n = 1, 041, 048$)											
Realized excess return	0.04	1.22	-1.14	-1.02	-1.00	-0.20	0.70	1.59	5.85	1.41	2.37
EEM-based expected return	1.69	3.99	-1.13	-0.98	-0.56	0.37	2.02	5.77	26.95	3.03	10.86
Option maturity	234.43	212.65	10.00	30.00	91.00	152.00	365.00	547.00	730.00	1.14	0.17
Option moneyness	1.00	0.12	0.41	0.85	0.94	1.00	1.06	1.13	1.97	0.02	2.57
Implied volatility	0.20	0.07	0.05	0.12	0.14	0.19	0.23	0.29	1.28	1.36	4.1
Underlying volatility	0.17	0.09	0.02	0.10	0.11	0.15	0.21	0.27	1.37	2.47	11.01
Option delta	0.51	0.24	0.10	0.15	0.30	0.50	0.70	0.85	0.90	-0.04	-1.18
C: Put Options:($n = 1, 067, 323$)											
Realized excess return	-0.49	1.01	-1.14	-1.02	-1.01	-1.00	-0.38	0.84	5.85	2.65	8.05
EEM-based expected return	0.05	1.65	-1.13	-1.00	-0.92	-0.51	0.38	1.72	26.85	4.13	29.27
Option maturity	241.72	217.00	10.00	30.00	91.00	152.00	365.00	547.00	730.00	1.07	-0.02
Option moneyness	1.01	0.13	0.51	0.86	0.95	1.01	1.07	1.16	3.05	0.84	4.76
Implied volatility	0.20	0.07	0.03	0.12	0.14	0.19	0.23	0.29	1.21	1.50	5.04
Underlying volatility	0.17	0.09	0.02	0.10	0.11	0.15	0.21	0.27	1.37	2.42	10.53
Option delta	-0.50	0.24	-0.90	-0.85	-0.70	-0.50	-0.30	-0.15	-0.10	0.01	-1.20

Notes: This table reports descriptive statistics for main variables over the period from 1998 to 2021. Panel A, B, and C report the statistics for all options, call options, and put options, respectively. Realized excess return is the realized holding-to-maturity option excess return defined by equation (5). The EEM-based expected return is the excess return calculated by using the EEM theory under the Black-Scholes model defined in equation (4). Option maturity is the number of calendar days until option expiration. Option moneyness is the ratio between the underlying index price and the option strike price. Implied volatility is the option's implied volatility provided by OptionMetrics. Option delta is the option's Black-Scholes delta value provided by OptionMetrics.

in future research to enhance the structural model's prediction on all types of options.

4.2. Predictability comparison

Table 3 shows the out-of-sample R^2 for the pooled testing sample from January 2003 through December 2021 using the ten machine learning methods, with and without EEM-based predictors. Nonlinear models routinely outperform the predictability uncovered by linear models for index option returns. The first row of Table 3 reports that with benchmark predictors, the R^2_{OS} for the OLS on the full sample is -3.74% . The R^2_{OS} for the other linear models on the full sample are all positive, ranging from 6.03% for PLS to 9.06% for GLM and outperforming a naive benchmark of predicting delta-hedged returns

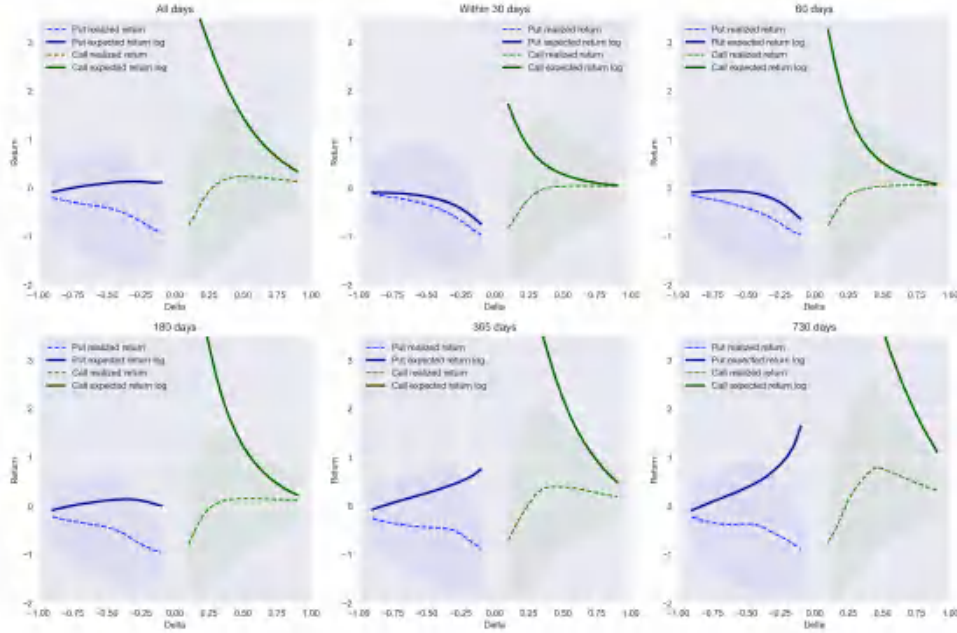


Figure 2: Realized returns and EEM-based expected returns of calls and puts

Notes: In this figure, dotted line displays the mean realized excess returns for calls ($\Delta > 0$) and puts ($\Delta < 0$) of various maturities in terms of delta values, along with the interval of one standard deviation in shaded area. Besides, we compare them with the average expected excess returns computed by the EEM theory.

of zero. With EEM-based determining predictors, OLS reduces 1.18% in R_{OS}^2 . With EEM-based resulting predictors, R_{OS}^2 is better, but still worse than the benchmark. It proved that OLS can't deal with multicollinearity effectively when adding more variables. While EEM-based resulting predictors have heteroplasmy compared to EEM-based determining predictors.

Table 3: Daily out-of-sample index option return prediction performance (R_{OS}^2 , %)

	OLS	PLS	PCR	LASSO	RIDGE	ENet	GLM	RF	GBDT	NN
With Benchmark predictors	-3.74	8.18	7.31	7.19	6.03	7.24	9.06	10.75	6.95	14.86
With EEM-based determining predictors	-4.92	6.35	7.12	7.64	6.85	7.46	8.36	12.17	10.93	7.56
With EEM-based resulting predictors	-4.60	6.44	7.98	7.83	7.20	7.89	8.65	12.19	11.34	16.17

Notes: This table shows R_{OS}^2 defined in equation (8) for ten models considered in full sample. NN1 outperforms other neural networks, hence the NN1 result is displayed in this table.

Linear model via dimension reduction improves predictions slightly weaker than the regularization model, but much better than the naive benchmark. By forming a few linear combinations of predictors, PCR and especially PLS, raise the out-of-sample R^2 to 7.31% and 8.18%, respectively. With EEM-based resulting predictors, PCR has a huge

improvement of 0.67% in R_{OS}^2 . Figure 3 and 4 show that PLS typically uses 1 to 13 components and PCR uses 2 to 13 components in forecasts with EEM-based resulting predictors.

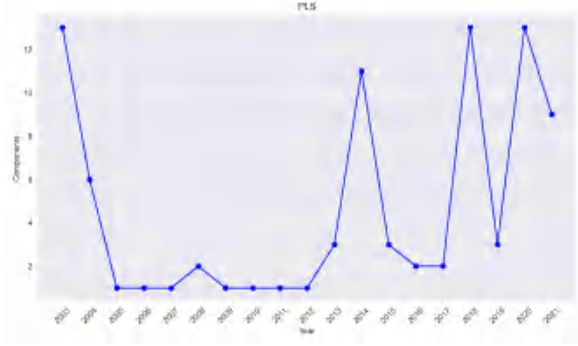


Figure 3: PLS regression components

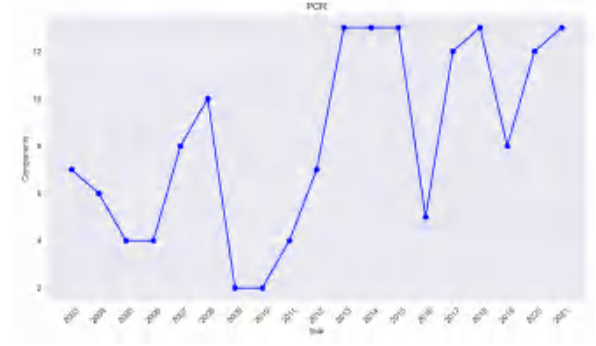


Figure 4: PCR regression components

Notes: This figure demonstrates the selected components of PLS and PCR in each training sample with EEM-based resulting predictors of our 19-year iteration out-of-sample analysis.

The regularization model improves predictions even further. With benchmark predictors, Elastic Net is the best regularization model with a R_{OS}^2 of 7.24%. It's the same with EEM-based resulting predictors. Elastic Net is a method that combines the functions of Lasso and ridge. While Ridge is worse than Lasso and Elastic Net, it still performs better than OLS.

With benchmark predictors, GLM shows the best performance over linear models with a R_{OS}^2 of 9.06%. The fact that this method uses spline functions of individual features, suggests that univariate expansions provide more incremental information beyond the linear model. With EEM-based resulting predictors, it decreases to 8.65%. More information can be extracted compared to EEM-based predictors. The extra variables indeed become the sources of multicollinearity, proving that GLM can't deal with multicollinearity effectively as well.

Our three nonlinear methods include RF (random forest), GBDT (gradient boosting decision tree), and NN (neural network). It's obvious that all of them perform much better with EEM-based predictors than linear methods. Among them, the neural network is the best. It has an R_{OS}^2 of 16.17% with EEM-based resulting predictors. It's a surprising result. GBDT performs badly, especially with Benchmark predictors. Since benchmark

predictors are not enough to construct strong learners used by boosting tree models in terms of quantity. With EEM-based predictors, it gets better. Random forest has an R_{OS}^2 of 10.75% with benchmark predictors, which is better than linear models' R_{OS}^2 . With EEM-based resulting predictors, RF has an increase of 1.44% in R_{OS}^2 .

Besides, note that EEM-based resulting predictors can improve the out-of-sample performance of most of the methods around full samples. This proves that the application of the structured model is successful in our study. Its constructed variables can bring incremental information.

Stable predictability is imperative for investors who wish to use the model forecasts in their investment decisions. Following [Bali et al. \(2023\)](#), we now provide estimates of the predictive power per year, which allows us to investigate the stability of the forecasts over time.

As in [Figure 5](#), contrasting to [Bali et al. \(2023\)](#), we find that in our study, though nonlinear models fit much better than the linear models in the training sample, but produce larger dispersion. Besides, it is worth noting that R_{IS}^2 of the neural network with EEM-based predictors is much better and more dispersed than that without EEM-based predictors. It's quite possible as a proof of incremental information found by the neural network.

In the testing sample, we can obviously see that only OLS produces a large dispersion. The median of OLS is close to 0, but the quartiles are less than -10%. In the year with the worst forecast, R_{OS}^2 reached a minimum of less than -50%.

Besides, we can see more intuitively that EEM-based predictors can improve the performance of most methods by improving in-sample fitness and out-of-sample prediction ability. Although the EEM-based predictors do not suppress the volatility of the R_{OS}^2 , they can raise the lower bound of it in most cases.

In [Figure 6](#) (a),(b), and (c) we show the detail. Amazingly, machine learning methods perform better than the benchmark in these situations in most years. The poorest performance of the OLS in 2009 could be attributed to the economic crisis of 2008 altering predictive trends because the Black-Scholes model under the EEM theory makes it

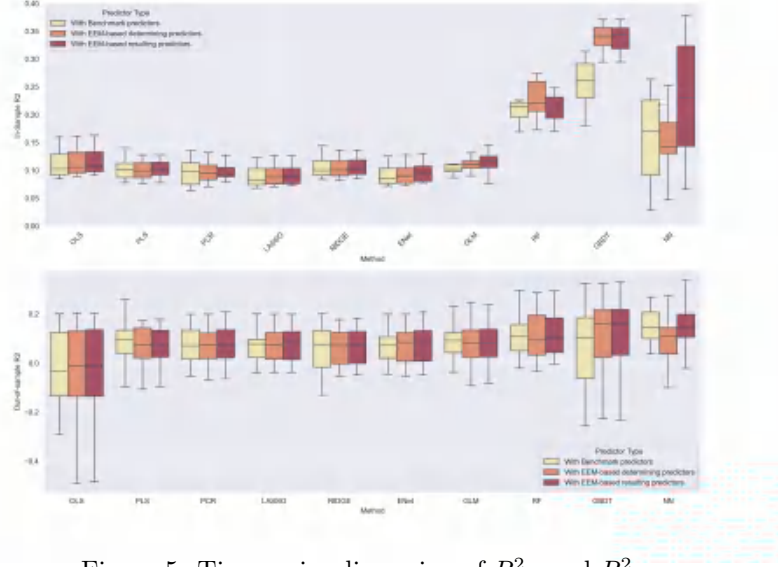


Figure 5: Time-series dispersion of R^2_{IS} and R^2_{OS}

Notes: This figure shows the dispersion of daily-based R^2_{IS} and R^2_{OS} defined in equation (8) in different machine learning methods. We show the 5th and 95th percentile R^2 in the whiskers, the interquartile range in the boxes, as well as the median (bar). NN shows the result of NN1 which is better than other neural networks.

difficult to provide effective information for this situation without jump risk parameters³. However, the predictive performance of machine learning methods remained above the baseline during that period, which can largely be attributed to the inherent advantages of the models themselves. In Figure 6 (d), we can also see the gap with and without the EEM-based resulting predictors. To our delight, R^2_{OS} are obviously better with EEM-based predictors than that of benchmark predictors during the economic crisis. Only in the PLS of 2004 and 2015, and random forest of 2009, 2013, 2016, and 2020, the predictive performance worsened after including the EEM-based predictors. Hence, we will consider jump risk in our models in our future study.

4.3. Which predictor matters?

We then analyze the importance of option characteristics. To begin, for each method, we refer to Gu et al. (2020) and calculate the reduction in R^2 from setting all values of a given predictor to zero within each training sample, and average these into a single

³This also proves the necessity of adopting the SV, SVJ, and SVJJ models under the EEM theory in our future research

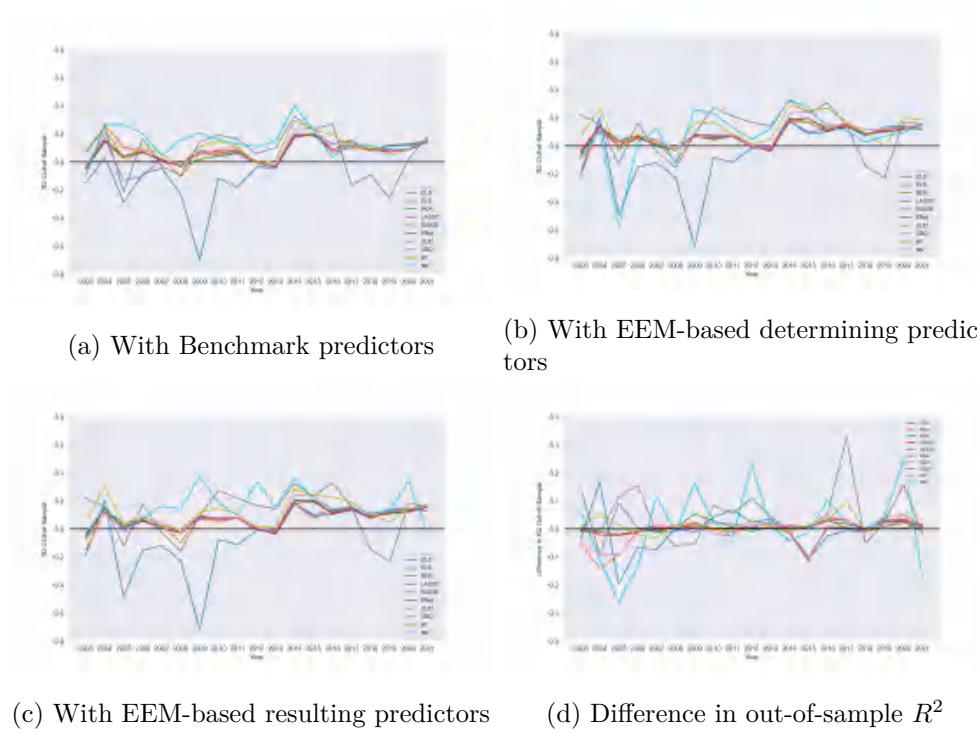


Figure 6: Time-series patterns of R^2_{OS}

Notes: This figure shows the daily-based R^2_{OS} with Benchmark predictors and with EEM-based predictors in different machine learning methods each year. At the same time, we also show the difference between R^2_{OS} with EEM-based resulting predictors and R^2_{OS} with benchmark predictors

importance measure for each predictor.

We find that our EEM-based predictors lay a certain role in models other than random forest. It indicates that both the linear form and the nonlinear form of the EEM-based predictors play an important role in the prediction of realized returns.

Surprisingly, net equity expansion is the most important variable in OLS, GBDT, and neural networks. In other methods, it is also in an important position. Since option returns are closely related to market expectations, if the market perceives the equity expansion positively, it could lead to a rise in stock prices, affecting the intrinsic value of options. Conversely, if the market views this as a dilution of share value, it could lead to a decrease in stock prices, again affecting option returns.

As a dimension reduction method, PCR is to find the latent factor with the largest variance. In its first step, PCA can use all variables to construct latent factors. They are grouped by reaching the biggest variance between latent factors. The selection of variables can influence the construction of latent factors.

We can see that just a few variables are important in PLS. Because PLS's purpose is to find latent factors related to predicted variables, but it can only find linear relationships, so some variables' relationships which may be nonlinear can't be found by the PLS method.

Both Lasso's sparse variable selection and Ridge's dense variable selection are shown in Figure 7. We can see just five variables are selected by Lasso, which are all our EEM-based predictors. In Ridge, our EEM-based predictors are also vital. Note that the least important variable is close to zero, but not equal to zero. This is the one feature of the Ridge method.

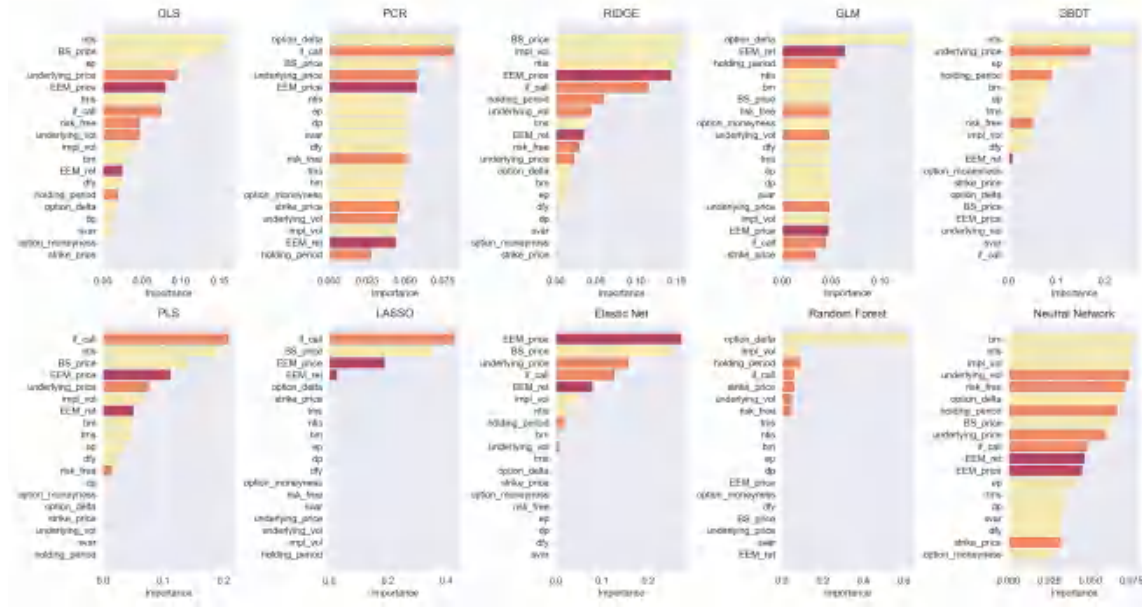


Figure 7: Variable importance by model

Notes: This figure shows variable importance for all 17 variables in each model. Benchmark predictors are in yellow. EEM-based determining predictors are in orange. EEM-based resulting predictors are in red. Variable importance is an average on all training samples. Variable importance within each model is normalized to sum to one.

Random forest selects a few variables in the nonlinear model. The possible reason should be that it is based on the set of weak learners which is hard to find deeper relationships, while GBDT is based on strong learners generated by residual iteration. In section 4.2, we can see that random forest have the smallest improvement of 0.03% after incorporating the EEM-based predictors.

Figure 8 reports the overall rankings of predictors for all models. We rank the importance of each predictor for each method and then sum their ranks. Predictors are

ordered so that the highest total ranks are on top and the lowest-ranking predictors are at the bottom. The color gradient within each column shows the model-specific ranking of predictors from least to most important (lightest to darkest).

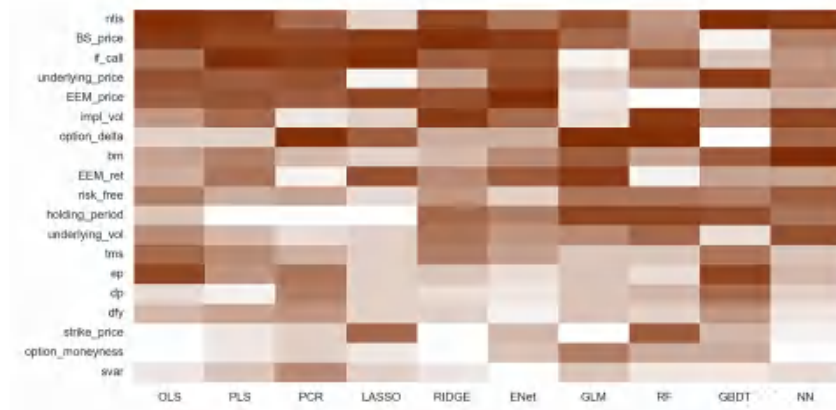


Figure 8: Characteristic importance

Notes: This figure shows rankings of EEM-based predictors and benchmark predictors in terms of overall model contribution. Characteristics are ordered based on the sum of their ranks over all models, with the most influential characteristics on the top and the least influential on the bottom. Columns correspond to the individual models, and the color gradients within each column indicate the most influential (dark brown) to the least influential (white) variables.

We can see that the EEM-based determining predictors are more important than the benchmark predictors. The EEM-based expected return, which is one of our EEM-based resulting predictors, is the fifth important variable. The EEM-based predictors occupy a high importance in such high dimension dataset.

5. Conclusion

This paper connects the economic theoretical finite horizon expected option return framework with machine learning in predicting future index option returns. We apply the equivalent expectation measure theory of [Nawalkha and Zhuo \(2022\)](#) to the Black-Scholes structural model, and construct the EEM-based predictors consisting of the theoretical determinants and predictions of expected option returns. Our findings underscore the substantial improvement in predictive accuracy achieved through the inclusion of EEM-based predictors across various machine learning models, particularly in scenarios marked by economic volatility or crisis periods. These predictors have consistently demonstrated

significant importance in enhancing model performance, with variable importance metrics exceeding zero across all tested models. Notably, our analysis reveals a clear superiority of nonlinear models over linear counterparts in forecasting S&P 500 index option returns, highlighting the efficacy of complex model architectures in capturing the intricacies of market dynamics.

In summary, the incorporation of EEM-based predictors within machine learning methodologies not only validates the practical applicability of the EEM theory but also confirms its important role in advancing financial predictive analytics. Looking forward, the application of the EEM theory to more sophisticated structural models, such as those incorporating stochastic volatility or affine jump-diffusion processes, promises to unveil even more profound insights. This prospective research could further enhance our understanding and prediction of option returns, solidifying the value of integrating economic theory with machine learning to tackle complex financial forecasting challenges.

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